

NF | DE-ESSER

USER MANUAL



Dynamic, transparent sibilance control

VST3 | AU | AAX Windows / macOS

Version 1.0.2 | NF Audio Tools by Nenno Fernando

Overview

NF De-Esser reduces excessive sibilance ("s", "sh", "ch") on vocals, instruments and mixes. A detector (side-chain) listens to the sound in real time and only turns it down when the sibilance goes past the limit you set. The rest of the voice is left untouched and its natural brightness is kept. There are just three main controls: Threshold, Range and Frequency.

SIGNAL FLOW



DYNAMIC, NOT A FILTER

The plug-in only acts when the "s" goes past the Threshold. Vowels and normal syllables stay at 0 dB of reduction.

1.5 MS LOOK-AHEAD

The reduction is ready when the "s" arrives, so the start of the sibilance is not clipped. The latency is reported to the host.

VOICE AND INSTRUMENTS

The VOICE/INSTR button adapts the detector: voice (protects the vowels) or instrument (guitar, cymbals, buses).

LISTEN

Hear exactly what the detector hears, to find the right Frequency with precision.

QUICK START

- 1 Pick the preset closest to your material in the preset tab (for example Default for voice, Acoustic Guitar for guitar).
- 2 Turn LISTEN on and turn Frequency until you hear the strongest "s". Go back to AUDIO.
- 3 Lower the Threshold until the plug-in catches the "s" sounds that bother you, watching the REDUCTION meter. Stop before it reacts to normal syllables.
- 4 Balance with Range: further right removes more of the "s"; further left lets the "s" through.

Interface map



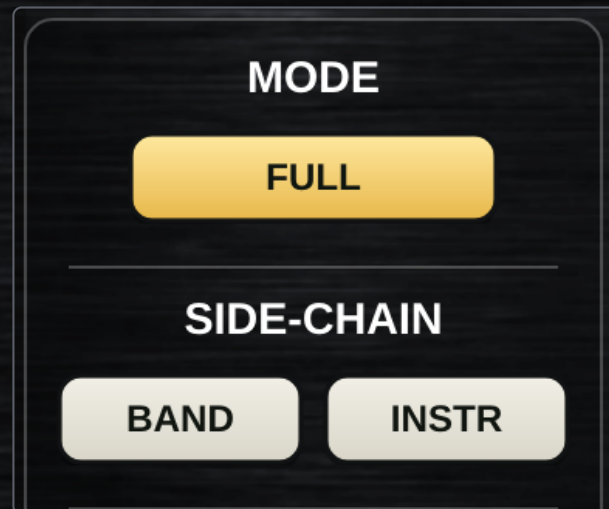
- 1 PRESET TAB**
Arrows step through the presets; click the name for the list, save or load.
- 2 MENU (3 LINES)**
Opens the manuals (Portuguese and English) and About.
- 3 POWER**
Turns the processing on and off. When off, the audio passes unchanged with the same delay.
- 4 MODE**
TARGET lowers only the highs; FULL lowers the whole signal.
- 5 SIDE-CHAIN**
HIGH/BAND picks the detector filter; VOICE/INSTR picks the kind of material.
- 6 RANGE**
Maximum reduction, from 0 to 20 dB.
- 7 FREQUENCY**
Region of the sibilance, from 500 Hz to 16 kHz.
- 8 MONITOR**
AUDIO plays the normal sound; LISTEN plays what the detector hears.
- 9 THRESHOLD**
Fader for the level where it starts to act, from -40 to 0 dB.
- 10 IN METER**
Level the detector hears, next to the Threshold.
- 11 REDUCTION**
How much the plug-in is turning the signal down, 0 to 20 dB.
- 12 L/R METERS**
Output level, 0 to -30 dB (read-only).
- 13 NF LOGO**
Click or double-click to return the window to its default size.

Mode: TARGET and FULL

The MODE button sets where the reduction is applied in the audio.



MODE on TARGET



MODE on FULL (with BAND and INSTR)

TARGET (DEFAULT)

Lowers only the high region, from about half the Frequency up. The body of the voice, the lows and the mids stay untouched. This is the most transparent mode.

FULL

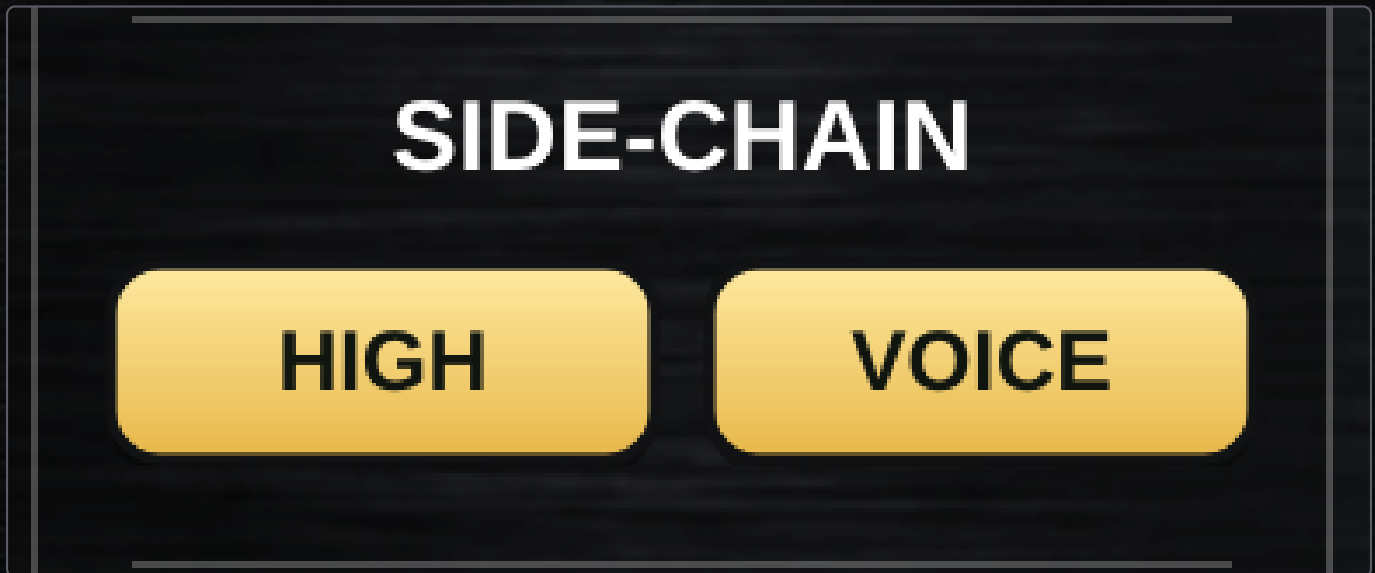
Lowers the whole signal, like a wideband compressor. It removes more "s" and suits material where the "s" dominates, but it also changes the body of the sound.

TIP

If the "s" still bothers you in TARGET, try lowering the Threshold or raising the Range before using FULL.

Side-Chain: filter and source

The side-chain is the part of the plug-in that listens to the sound to decide when to reduce. This section has two buttons side by side.



SIDE-CHAIN section: HIGH/BAND and VOICE/INSTR

HIGH (DEFAULT)

The detector listens to everything above the Frequency (4th-order high-pass filter). It removes the most "s" and works well on voices with several kinds of "s", guitar and cymbals.

BAND

The detector listens to a narrow band around the Frequency. It isolates one specific kind of "s" but reduces less. Use it for fine tuning.

VOICE (DEFAULT)

The reduction only comes in where the highs stand out from the body of the sound, that is, on an "s" and not on a bright vowel. It protects the tone of the voice.

INSTR

Level-only detection, without the sibilance filter. Needed on pick guitar, cymbals and buses, where the body of the sound always outweighs the highs.

Range

Range is the ceiling of the reduction, from 0 to 20 dB. It is the fine-tuning control for the balance.



RANGE knob (0 to 20 dB)

TURN RIGHT: MORE REDUCTION

The plug-in can take more of the "s". Go too far and the "s" gets held back, as if the singer's tongue were stuck.

TURN LEFT: LESS REDUCTION

The "s" sounds more open and natural again. At 0 dB the plug-in reduces nothing.

The plug-in only reduces what the Threshold lets through, up to the Range limit. Double-clicking the knob returns it to its default value.

Frequency

Frequency sets the region where you hear the sibilance, from 500 Hz to 16 kHz. The knob is logarithmic: the middle of the scale sits around 3 kHz, where there is more precision.



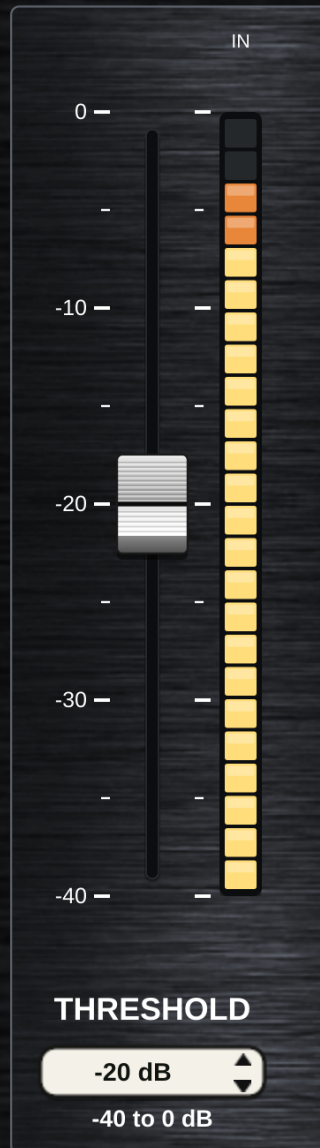
FREQUENCY knob (500 Hz to 16 kHz)

Male voice	Usually between 3 and 5 kHz.
Female voice	Usually between 5 and 8 kHz.
Guitar and cymbals	From 7 to 9 kHz or more.

These values are starting points. Use LISTEN to hear what the detector hears and look for the region where the "s" bothers you most. Frequency glides smoothly as you turn it, without clicks.

Threshold

Threshold is the large fader: it sets the level where the plug-in starts to act, from -40 to 0 dB. The IN meter sits beside it.



THRESHOLD fader and IN meter

LOWER

The plug-in catches more "s" sounds, including the soft ones.

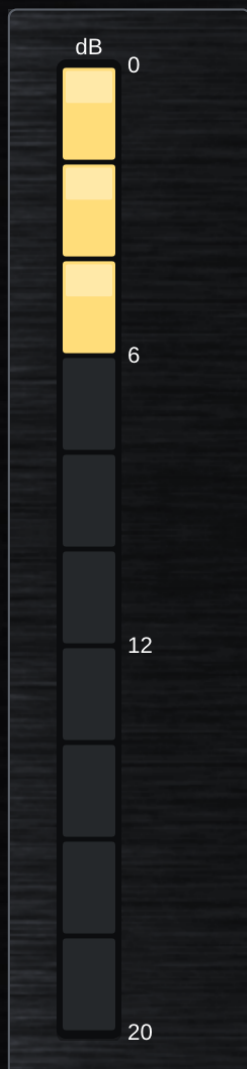
HIGHER

The plug-in only acts on the strongest "s" sounds. At 0 dB there is no reduction.

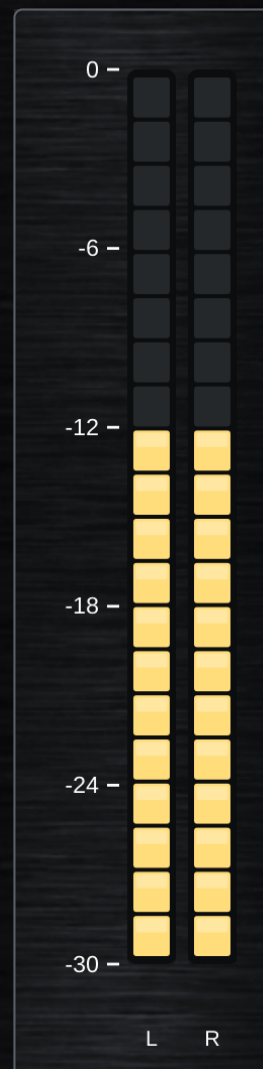
The IN meter shows the level the detector hears, on the same scale as the fader. When the level goes past the fader position, the plug-in starts to reduce. The scale is calibrated so that a typical "s" sits in the upper part.

Meters

Three groups of meters show what is going on. All of them are read-only.



REDUCTION: 0 to 20 dB



Output meters L and R: 0 to -30 dB

REDUCTION

Shows how much the plug-in is turning the signal down at that moment, in dB. If it barely lights up, the Threshold is too high or there is no sibilance.

L / R

Output level of each channel. The De-Esser has no output control because it only lowers the signal, with no make-up gain.

Monitor: AUDIO and LISTEN

MONITOR chooses what you hear at the plug-in output.



AUDIO (normal sound)



LISTEN (side-chain)

AUDIO

Plays the processed audio, with the reduction applied.

LISTEN

Plays what the detector hears, that is, the sound filtered by HIGH or BAND around the Frequency. Use it to find the right Frequency and to check that the detector only catches the "s".

NOTE

In LISTEN, Threshold and Range do not change what you hear, only Frequency and the filter do. When you turn Frequency while the voice plays, the sound varies a lot, because dry consonants appear and disappear as the cutoff passes through them. This is normal.

Presets, Power and window

The top of the plug-in holds the presets, the menu, the Power button and the logo.



Preset tab, menu and Power



NF logo

PRESET TAB

The arrows step through the presets. Click the name to open the list with the 13 factory presets, plus Save Preset and Load Preset. Saved presets live in Documents/NF Audio Tools/NF De-Esser/Presets.

POWER

When off, the audio passes unprocessed with the same delay as the plug-in, so the timing does not change when you toggle it.

MENU (3 LINES)

Opens both PDF manuals (Portuguese and English) in the computer's default viewer, and About, with the plug-in version.

WINDOW SIZE

The window is square. Drag the corner to resize it, from 405 to 1200 px. The size is remembered when you reopen it. Click or double-click the NF logo to return to the default (540 px).

Factory presets

The plug-in comes with 13 factory presets. Each one loads Frequency, Threshold, Range, the HIGH/BAND filter and the VOICE/INSTR source. The first eight are for voice and open on VOICE; the rest are for instruments and buses and open on INSTR. Presets are starting points: adjust them to your material.

PRESET	FREQUENCY	THRESHOLD	RANGE	FILTER	SOURCE
Default	5.5 kHz	-20 dB	12 dB	HIGH	VOICE
Vocal Male	4.5 kHz	-18 dB	10 dB	BAND	VOICE
Vocal Male Sh	3.4 kHz	-18 dB	8 dB	BAND	VOICE
Vocal Female	6.8 kHz	-18 dB	10 dB	BAND	VOICE
Vocal Female Sh	5.1 kHz	-18 dB	8 dB	BAND	VOICE
Vocal Wide	5.0 kHz	-20 dB	10 dB	HIGH	VOICE
Vocal Rock	5.5 kHz	-20 dB	12 dB	HIGH	VOICE
Vocal Podcast	4.5 kHz	-16 dB	8 dB	HIGH	VOICE
Acoustic Guitar	7.5 kHz	-16 dB	6 dB	HIGH	INSTR
Hi-Hat Tamer	9.0 kHz	-16 dB	8 dB	HIGH	INSTR
Overheads	8.5 kHz	-18 dB	6 dB	HIGH	INSTR
Mix Bus	6.5 kHz	-12 dB	4 dB	HIGH	INSTR
Master Gentle	7.5 kHz	-10 dB	3 dB	HIGH	INSTR

Presets you save appear next to the factory ones and keep every control, including VOICE/INSTR.

How to set it up

A simple routine to get the plug-in right on your material.

- 1 Choose the Frequency: turn LISTEN on and look for where the "s" is strongest. Go back to AUDIO.
- 2 Lower the Threshold to where you feel it sounds good, watching REDUCTION. It should light up on the "s" sounds and not on normal syllables.
- 3 Adjust Range to balance. Further right holds the "s" back; further left lets it through.
- 4 If sibilance is still left, try FULL or lower the Threshold. If the "s" gets dull, lower the Range or raise the Threshold.

BY MATERIAL

VOICE

Use VOICE and HIGH. Start from the Default or a Vocal preset. Usually 6 to 10 dB of maximum reduction on the strongest "s" sounds is enough.

PICK GUITAR

Use INSTR and HIGH, Acoustic Guitar preset. If it acts too little, lower the Threshold.

CYMBALS AND OVERHEADS

Use INSTR, Hi-Hat Tamer or Overheads preset, Frequency between 8 and 9 kHz.

MIX BUS AND MASTER

Use INSTR, Mix Bus or Master Gentle preset, with a low Range (3 to 4 dB) for a light touch.

Specifications and troubleshooting

Technical data and answers to common questions.

Formats	VST3, AU and AAX.
Channels	Mono and stereo, with stereo-linked detection.
Frequency	500 Hz to 16 kHz.
Threshold	-40 to 0 dB.
Range	0 to 20 dB.
Latency	1.5 ms look-ahead (72 samples at 48 kHz), reported to the host for automatic compensation.
Attack / release	0.4 ms attack; automatic release from 12 to 24 ms, faster on short "s" sounds and smoother on long ones.

FREQUENTLY ASKED QUESTIONS

THE PLUG-IN REDUCES NOTHING

The Threshold may be too high, Range at 0 or Power off. On guitar, check that it is on INSTR.

THE "S" GOT DULL

Lower the Range, raise the Threshold or use BAND. Check in LISTEN that the Frequency is in the right place.

CLICK WHEN TURNING FREQUENCY IN LISTEN

The voice has dry consonants that appear and disappear as the cutoff passes through them. This is expected in LISTEN.

THE PLUG-IN IS SILENT

Check that the plug-in is activated (license).